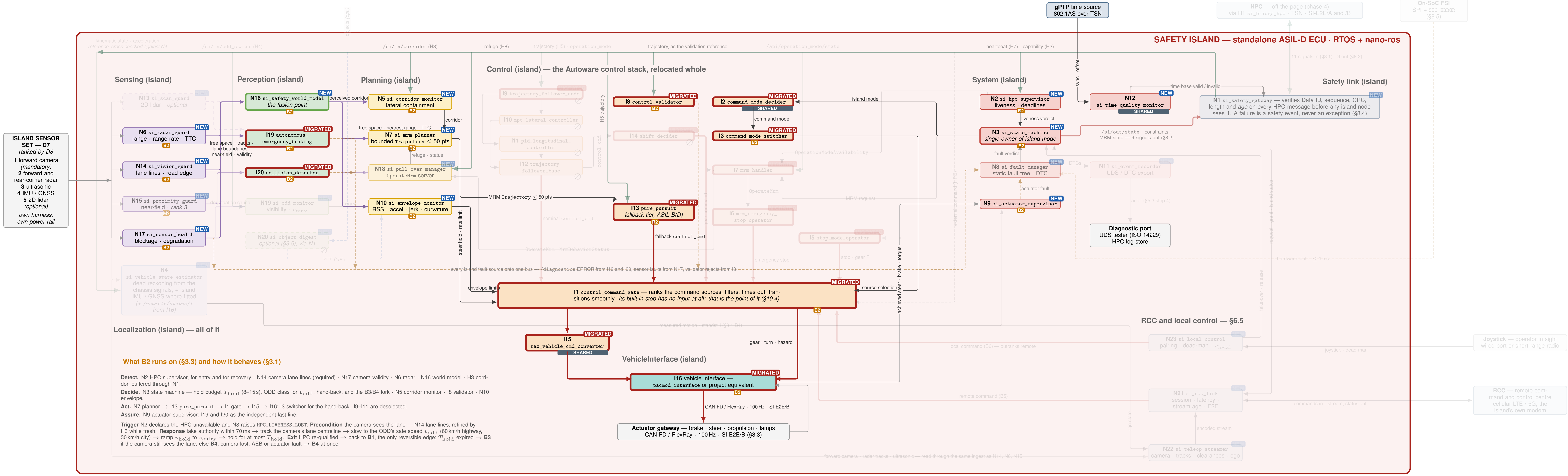
Phase 5 · B2 — Keep the vehicle in its lane after loss of the HPC

When the HPC is unavailable and B1 therefore cannot be used, take authority and hold the vehicle in its current lane — for a bounded 8–15 s window, first down to the ODD's safe speed — while the camera still sees the lane.
The phase-5 drawing, unchanged: what B2 runs on — and the SHARED nodes every behaviour runs through — is in full colour, what it merely passes through is half-tone, and the rest of the island is faded.


gPTP time source
802.1AS over TSN
HPC — off the page (phase 4)
via H1 si_bridge_hpc · TSN · SI-E2E/A and /B
On-SoC FSI
SPI + SOC_ERROR
(§8.5)
SAFETY ISLAND — standalone ASIL-D ECU · RTOS + nano-ros
objects (opt.)
kinematic state · acceleration
reference, cross-checked against N4
/si/in/odd_status (H4)
/si/in/corridor (H3)
refuge (H8)
trajectory (H5) · operation_mode
trajectory, as the validation reference
/api/operation_mode/state
heartbeat (H7) · capability (H2)
sync · offset
11 signals in (§8.1) · 9 out (§8.2)
ISLAND SENSOR SET — D7
ranked by D8
1 forward camera
(mandatory)
2 forward and rear-corner radar
3 ultrasonic
4 IMU / GNSS
5 2D lidar
(optional)
own harness, own power rail
Sensing (island)
N13 si_scan_guard
2D lidar · optional
NEW
N6 si_radar_guard
range · range-rate · TTC
B2
N14 si_vision_guard
lane lines · road edge
N15 si_proximity_guard
near-field · rank 3
N17 si_sensor_health
blockage · degradation
N4
si_vehicle_state_estimator
dead reckoning from the chassis signals, + island IMU / GNSS where fitted
(+ /vehicle/status/* from I16)
Localization (island) — all of it
Perception (island)
N16 si_safety_world_model
the fusion point
MIGRATED
I19 autonomous_emergency_braking
I20 collision_detector
free space · tracks · lane boundaries · near-field · validity
degradation cause
N19 si_odd_monitor
visibility · $v_{\max}$
N20 si_object_digest
optional (§3.5), via N1
veto (opt.)
Planning (island)
perceived corridor
N5 si_corridor_monitor
lateral containment
corridor
free space · nearest range · TTC
N7 si_mrm_planner
bounded Trajectory ≤ 50 pts
refuge · status
N18 si_pull_over_manager
OperateMrm server
$v_{\max}$
N10 si_envelope_monitor
RSS · accel · jerk · curvature
MRM Trajectory ≤ 50 pts
steer hold · rate limit
envelope limits
Control (island) — the Autoware control stack, relocated whole
I9 trajectory_follower_node
I10 mpc_lateral_controller
I11 pid_longitudinal_controller
I12 trajectory_follower_base
control_cmd
H5 trajectory
I8 control_validator
I14 shift_decider
I13 pure_pursuit
fallback tier, ASIL-B(D)
nominal control_cmd
fallback control_cmd
gear command
OperateMrm · MrmBehaviorStatus
I2 command_mode_decider
SHARED
command mode
I3 command_mode_switcher
island mode
OperationModeAvailability
I7 mrm_handler
OperateMrm
MRM request
I6 mrm_emergency_stop_operator
I5 stop_mode_operator
emergency stop
stop · gear P
external command (HPC) †
every island fault source onto one bus — /diagnostics ERROR from I19 and I20, sensor faults from N17, validator rejects from I8
I1 control_command_gate — ranks the command sources, filters, times out, transitions smoothly. Its built-in stop has no input at all: that is the point of it (§10.4).
source selection
achieved steer · brake · torque
measured motion · standstill (§3.1 B4)
I15
raw_vehicle_cmd_converter
gear · turn · hazard
VehicleInterface (island)
I16 vehicle interface —
pacmod_interface or project equivalent
CAN FD / FlexRay · 100 Hz · SI-E2E/B
Actuator gateway — brake · steer · propulsion · lamps
CAN FD / FlexRay · 100 Hz · SI-E2E/B (§8.3)
System (island)
N2 si_hpc_supervisor
liveness · deadlines
liveness verdict
N3 si_state_machine
single owner of island mode
fault verdict
N8 si_fault_manager
static fault tree · DTC
DTCs
actuator fault
N9 si_actuator_supervisor
N12
si_time_quality_monitor
time base valid / invalid
N11 si_event_recorder
UDS / DTC export
audit (§5.3 step 4)
Diagnostic port
UDS tester (ISO 14229)
HPC log store
/si/out/state · constraints ·
MRM state — 9 signals out (§8.2)
Safety link (island)
N1 si_safety_gateway — verifies Data ID, sequence, CRC, length and age on every HPC message before any island node sees it. A failure is a safety event, never an exception (§8.4)
request · grant · island status
take-over · release
hardware fault · ≤ 1 ms
RCC and local control — §6.5
N23 si_local_control
pairing · dead-man · v_{local}
local command (B6) — outranks remote
joystick · dead-man
Joystick — operator in sight
wired port or short-range radio
N21 si_rcc_link
session · latency · stream age · E2E
remote command (B5)
commands in · stream, status out
RCC — remote command and control centre
cellular LTE / 5G, the island's own modem
ego state
encoded stream
N22 si_teleop_streamer
camera · tracks · clearances · ego
forward camera · radar tracks · ultrasonic — read through the same ingest as N14, N6, N15

What B2 runs on (§3.3) and how it behaves (§3.1)

Detect. N2 HPC supervisor, for entry and for recovery · N14 camera lane lines (required) · N17 camera validity · N6 radar · N16 world model · H3 corridor, buffered through N1.

Decide. N3 state machine — hold budget T_{hold} (8–15 s), ODD class for v_{odd}, hand-back, and the B3/B4 fork · N5 corridor monitor · I8 validator · N10 envelope.

Act. N7 planner → I13 `pure_pursuit` → I1 gate → I15 → I16; I3 switcher for the hand-back. I9–I11 are deselected.

Assure. N9 actuator supervisor; I19 and I20 as the independent last line.

Trigger N2 declares the HPC unavailable and N8 raises `HPC_LIVENESS_LOST`. **Precondition** the camera sees the lane — N14 lane lines, refined by H3 while fresh. **Response** take authority within 70 ms → track the camera's lane centreline → slow to the ODD's safe speed v_{odd} (60 km/h highway, 30 km/h city) → ramp v_{hold} to v_{entry} → hold for at most T_{hold}. **Exit** HPC re-qualified → back to **B1**, the only reversible edge; T_{hold} expired → **B3** if the camera still sees the lane, else **B4**; camera lost, AEB or actuator fault → **B4** at once.

Stacks — the colours of `ArchitectureMain.pdf`

Sensing · Localization · Perception · Planning · Control
System · VehicleInterface · Vehicle · Safety link

Where each node came from

MIGRATED — the same Autoware node, migrated
NEW — new — does not exist in Autoware

18 migrated + 19 new = **37**, plus the optional N20 — and N21–N23 of §6.5, outside the counts until their detailed design is done. Not drawn: I4, a header-only library with no topics, and I17 + I18, the legacy alternative of §4.3. I10 and I11 are plugins inside I9 and publish no topic of their own. † the HPC's external command reaches the gate (§5.2, §10.4) but has no row in the §8.1 table; B5's remote command arrives through N21 instead.

Behaviours each node enables (§3.3)

one fixed slot per behaviour B2 — **B1** degrade the ADS · **B2** lane-keep hold after HPC loss · **B3** pull over · **B4** blind emergency stop · **B5** tele-operation · **B6** joystick. A filled tag means the node appears in that behaviour's Detect, Decide, Act or Assure cell of §3.3.

SHARED — **SHARED** — every behaviour runs through it, so §3.3 names it against none: I2 turns island mode into command mode, I15 converts whichever command wins the gate, N12 qualifies the time base behind every age and deadline check.

B2 — **Hollow tag** — not in §3.3, but §3.1 puts the node on that behaviour's path: N15 feeds I20, B2's independent last line, and is the near-field check behind B3's success criterion.

∅ — **Used by no behaviour** — I9–I12 and I14 track the HPC's nominal trajectory, which B2–B4 deselect and B1 only constrains. N20 is optional (§3.5) and required by none.

Messages

─▶	HPC-sourced, verified by N1	─▶	actuator feedback
─▶	island sensor data	─▶	state · mode · decision
─▶	control command into the gate	- - -▶	optional (§3.5)
- - -▶	fault · `/diagnostics`		

Every edge on this page, with the design-document section it is taken from, is catalogued in `../messageflow/`, which draws the same 37 nodes coloured by execution partition instead.

In this view — B2, lane-keep hold

🟥 **full colour** — named for B2 in §3.3 (Detect · Decide · Act · Assure), or SHARED (I2, I15, N12)

▨ **half-tone** — on B2's path by §3.1, but not named in §3.3

☐ **faded** — not used by B2. An edge is drawn at the weaker of its two ends. Only B2's own tag is drawn on each box.